We use rectangular integration with a step size of 1 msec to simulate the system, the extended Kalman filter, and the unscented Kalman filter for 30 seconds. Figure 14.4 shows the altitude and velocity of the falling body. For the first few seconds, the velocity is constant. But then the air density increases and drag slows the falling object. Toward the end of the simulation, the object has reached a constant terminal velocity as the acceleration due to gravity is canceled by drag.

Figure 14.5 shows typical EKF and UKF estimation-error magnitudes for this system. It is seen that the altitude and velocity estimates both spike around 10 seconds, at which point the altitude of the measuring device and the falling body are about the same, so the measurement gives less information about the body's altitude and velocity. It is seen from the figure that the UKF consistently gives estimates that are one or two orders of magnitude better than the EKF.

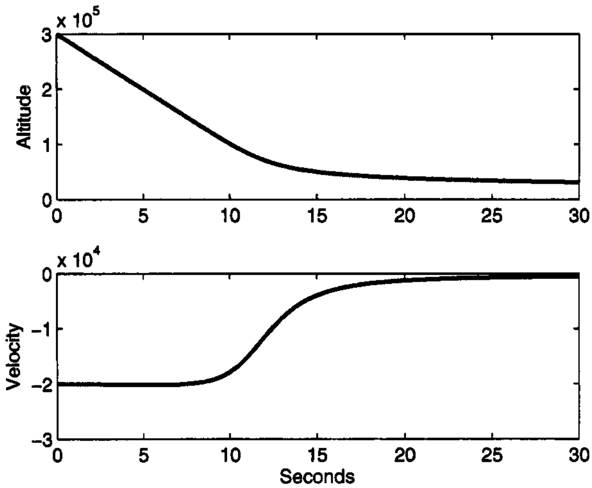

Figure 14.4 Altitude and velocity of a falling body for Example 14.2.

▽▽▽

14.4 OTHER UNSCENTED TRANSFORMATIONS

The unscented transformation discussed in the previous section is not the only one that exists. In this section, we discuss several other possible transformations. These other transformations can be used if we have some information about the statistics of the noise, or if we are interested in computational savings.

14.4.1 General unscented transformations

We have seen that an accurate mean and covariance approximation for a nonlinear transformation $y = h(x)$ can be obtained by choosing $2n$ sigma points (where n is the dimension of x) as given in Equation (14.29), and approximating the mean and

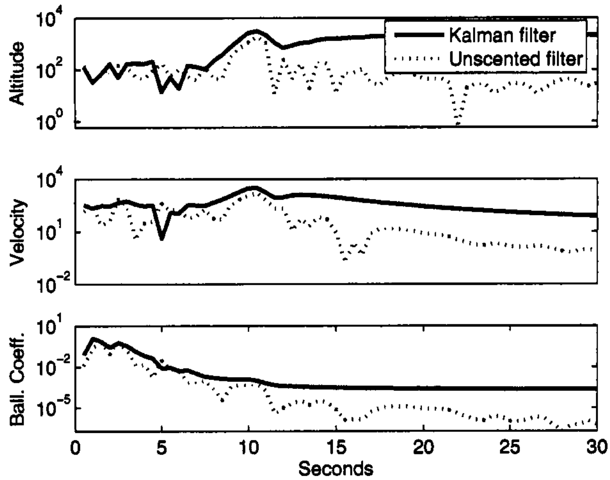

Figure 14.5 Kalman filter and unscented filter estimation-error magnitudes of the altitude, velocity, and ballistic coefficient of a falling body for Example 14.2.

covariance as given in Equations (14.33) and (14.43). However, it can be shown that the same order of mean and covariance estimation accuracy can be obtained by choosing $(2n+1)$ sigma points $x^{(i)}$ as follows:

$$\begin{aligned}
x^{(0)} &= \bar{x} \\
x^{(i)} &= \bar{x} + \tilde{x}^{(i)} \qquad i = 1, \cdots, 2n \\
\tilde{x}^{(i)} &= \left(\sqrt{(n+\kappa)P}\right)_i^T \qquad i = 1, \cdots, n \\
\tilde{x}^{(n+i)} &= -\left(\sqrt{(n+\kappa)P}\right)_i^T \qquad i = 1, \cdots, n
\end{aligned} \tag{14.74}$$

The $(2n+1)$ weighting coefficients are given as

$$\begin{aligned}
W^{(0)} &= \frac{\kappa}{n+\kappa} \\
W^{(i)} &= \frac{1}{2(n+\kappa)} \qquad i = 1, \cdots, 2n
\end{aligned} \tag{14.75}$$

The unscented mean and covariance approximations are computed as

$$\begin{aligned}
y^{(i)} &= h\left(x^{(i)}\right) \\
\bar{y}_u &= \sum_{i=0}^{2n} W^{(i)} y^{(i)} \\
P_u &= \sum_{i=0}^{2n} W^{(i)} \left(y^{(i)} - y_u\right)\left(y^{(i)} - y_u\right)^T
\end{aligned} \tag{14.76}$$

It can be seen that if $\kappa = 0$ then these definitions reduce to the quantities given in Section 14.2. Any κ value can be used [as long as $(n+\kappa) \neq 0$] and will give a mean and covariance estimation accuracy with the same order of accuracy as derived in Section 14.2. However, κ can be used to reduce the higher-order errors of the mean and covariance approximation. For example, if x is Gaussian then $\kappa = 3 - n$ will minimize some of the errors in the fourth-order terms in the mean and covariance approximation [Jul00, Jul04].

14.4.2 The simplex unscented transformation

If computational effort is a primary consideration, then a minimum number of sigma points can be chosen to give the order of estimation accuracy derived in the previous section. It can be shown [Jul02a, Jul04] that if x has n elements then the minimum number of sigma points that gives the order of estimation accuracy of the previous section is equal to $(n+1)$. These sigma points are called simplex sigma points. The following algorithm results in $(n+2)$ sigma points, but the number can be reduced to $(n+1)$ by choosing one of the weights to be zero. The simplex sigma-point algorithm can be summarized as follows.

The simplex sigma-point algorithm

1. Choose the weight $W^{(0)} \in [0, 1)$. The choice of $W^{(0)}$ affects only the fourth and higher order moments of the set of sigma points [Jul00, Jul02a].

2. Choose the rest of the weights as follows:

$$W^{(i)} = \begin{cases} 2^{-n}(1 - W^{(0)}) & i = 1, 2 \\ 2^{i-2} W^{(1)} & i = 3, \cdots, n+1 \end{cases} \tag{14.77}$$

3. Initialize the following one-element vectors:

$$\begin{aligned} \sigma_0^{(1)} &= 0 \\ \sigma_1^{(1)} &= \frac{-1}{\sqrt{2W^{(1)}}} \\ \sigma_2^{(1)} &= \frac{1}{\sqrt{2W^{(1)}}} \end{aligned} \tag{14.78}$$

4. Recursively expand the σ vectors by performing the following steps for $j = 2, \cdots, n$:

$$\sigma_i^{(j)} = \begin{cases} \begin{bmatrix} \sigma_0^{(j-1)} \\ 0 \end{bmatrix} & i = 0 \\ \begin{bmatrix} \sigma_i^{(j-1)} \\ \frac{-1}{\sqrt{2W^{(j+1)}}} \end{bmatrix} & i = 1, \cdots, j \\ \begin{bmatrix} 0_{j-1} \\ \frac{j}{\sqrt{2W^{(j+1)}}} \end{bmatrix} & i = j+1 \end{cases} \tag{14.79}$$

where 0_j is the column vector containing j zeros.

5. After the above recursion is complete we have the n-element vectors $\sigma_i^{(n)}$ ($i = 0, \cdots, n+1$). We modify the unscented transformation of Equation (14.29) and obtain the sigma points for the unscented transformation as follows:

$$x^{(i)} = \bar{x} + \sqrt{P}\sigma_i^{(n)} \quad (i = 0, \cdots, n+1) \tag{14.80}$$

We actually have $(n+2)$ sigma points instead of the $(n+1)$ sigma points as we claimed, but if we choose $W^{(0)} = 0$ then the $x^{(0)}$ sigma point can be ignored in the ensuing unscented transformation. The unscented Kalman filter algorithm in Section 14.3 is then modified in the obvious way based on this minimal set of sigma points.

The problem with the simplex UKF is that the ratio of $W^{(n)}$ to $W^{(1)}$ is equal to 2^{n-2}, where n is the dimension of the state vector x. As the dimension of the state increases, this ratio increases and can quickly cause numerical problems. The only reason for using the simplex UKF is the computational savings, and computational savings is an issue only for problems of high dimension (in general). This makes the simplex UKF of limited utility and leads to the spherical unscented transformation in the following section.

14.4.3 The spherical unscented transformation

The unscented transformation discussed in Section 14.2 is numerically stable. However, it requires $2n$ sigma points and may be too computationally expensive for some applications. The simplex unscented transformation discussed in Section 14.4.2 is the cheapest computational unscented transformation but loses numerical stability for problems with a moderately large number of dimensions. The spherical unscented transformation was developed with the goal of rearranging the sigma points of the simplex algorithm in order to obtain better numerical stability [Jul03, Jul04]. The spherical sigma points are chosen with the following algorithm.

The spherical sigma-point algorithm

1. Choose the weight $W^{(0)} \in [0, 1)$. The choice of $W^{(0)}$ affects only the fourth- and higher-order moments of the set of sigma points [Jul00, Jul02a].

2. Choose the rest of the weights as follows:

$$W^{(i)} = \frac{1 - W^{(0)}}{n+1} \quad i = 1, \cdots, n+1 \tag{14.81}$$

Note that (in contrast to the simplex unscented transformation) all of the weights are identical except for $W^{(0)}$.

3. Initialize the following one-element vectors:

$$\begin{aligned} \sigma_0^{(1)} &= 0 \\ \sigma_1^{(1)} &= \frac{-1}{\sqrt{2W^{(1)}}} \\ \sigma_2^{(1)} &= \frac{1}{\sqrt{2W^{(1)}}} \end{aligned} \tag{14.82}$$

4. Recursively expand the σ vectors by performing the following steps for $j = 2, \cdots, n$:

$$\sigma_i^{(j)} = \begin{cases} \begin{bmatrix} \sigma_0^{(j-1)} \\ 0 \end{bmatrix} & i = 0 \\ \begin{bmatrix} \sigma_i^{(j-1)} \\ \frac{-1}{\sqrt{j(j+1)W^{(1)}}} \end{bmatrix} & i = 1, \cdots, j \\ \begin{bmatrix} 0_{j-1} \\ \frac{j}{\sqrt{j(j+1)W^{(1)}}} \end{bmatrix} & i = j+1 \end{cases} \tag{14.83}$$

where 0_j is the column vector containing j zeros.

5. After the above recursion is complete, we have the n-element vectors $\sigma_i^{(n)}$ $(i = 0, \cdots, n+1)$. As with the simplex sigma points, we actually have $(n+2)$ sigma points above, but if we choose $W^{(0)} = 0$ then the $x^{(0)}$ sigma point can be ignored in the ensuing unscented transformation. We modify the unscented transformation of Equation (14.29) and obtain the sigma points for the unscented transformation as follows:

$$x^{(i)} = \bar{x} + \sqrt{P}\sigma_i^{(n)} \quad (i = 0, \cdots, n+1) \tag{14.84}$$

The unscented Kalman filter algorithm in Section 14.3 is then modified in the obvious way based on this set of sigma points.

The ratio of the largest element of $\sigma_i^{(n)}$ to the smallest element is

$$\frac{n}{\sqrt{n(n+1)W^{(1)}}} \Bigg/ \frac{1}{\sqrt{n(n+1)W^{(1)}}} = n \tag{14.85}$$

so numerical problems should not be an issue for the spherical unscented transformation.

■ EXAMPLE 14.3

Here we consider the falling-body system described in Example 14.2. The initial conditions of the system and the estimator are given as

$$\begin{aligned} x_0 &= \begin{bmatrix} 300,000 & -20,000 & 1/1000 \end{bmatrix}^T \\ \hat{x}_0^+ &= \begin{bmatrix} 303,000 & -20,200 & 1/1010 \end{bmatrix}^T \\ P_0^+ &= \begin{bmatrix} 30,000 & 0 & 0 \\ 0 & 2,000 & 0 \\ 0 & 0 & 1/10,000 \end{bmatrix} \end{aligned} \tag{14.86}$$

We ran 100 Monte Carlo simulations, each with a 60 s simulation time. The average RMS estimation errors of the EKF, standard UKF (six sigma points), simplex UKF (four sigma points since we chose $W^{(0)} = 0$), and spherical UKF (four sigma points since we chose $W^{(0)} = 0$) are given in Table 14.1. The simplex UKF performs best for altitude estimation, with the standard UKF